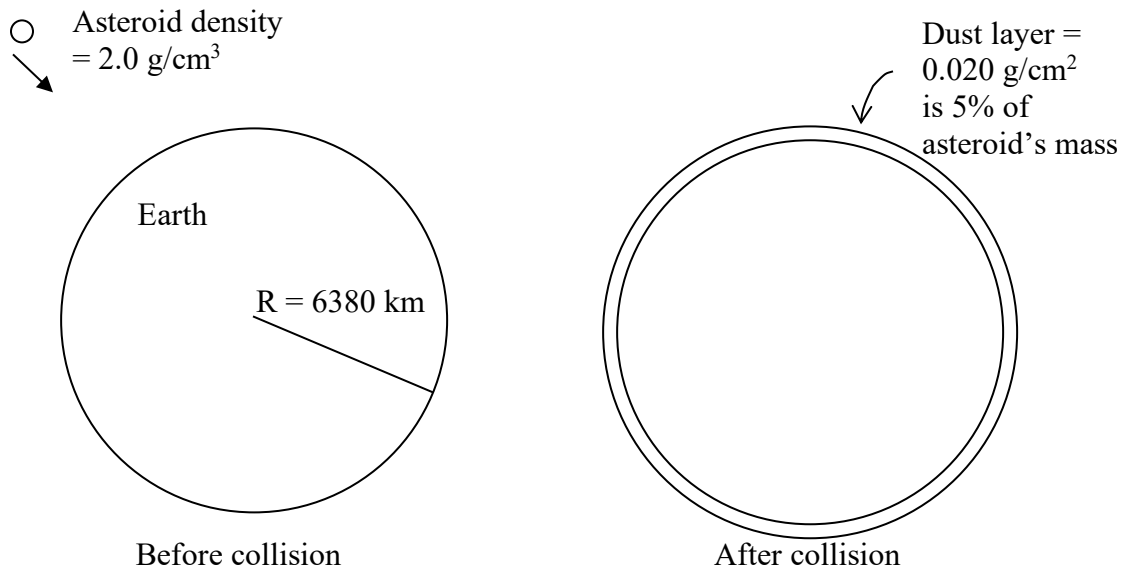


Solution to the Asteroid Problem: Group Problem 1

First, draw a useful picture:



Second, state the question that you need to answer:

Question: What was the radius of the asteroid?

Third, conceptually outline how you might answer this question:

Approach: All the dust layer comes from the asteroid.

5% of the mass of the asteroid is the dust layer on the Earth.

Assume that the thickness of the dust is small compared to the radius of the Earth.

First find the mass of the dust layer on the Earth using the surface area of a sphere.

Approximate the shape of the Earth as a sphere.

Then find the radius of the asteroid from its mass and density.

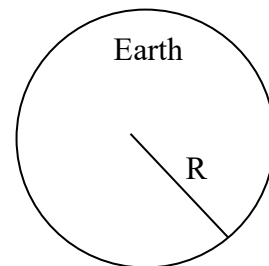
Approximate the shape of the asteroid as a sphere.



Mass of Asteroid = M

Volume of Asteroid = $V = \frac{4}{3}\pi r^3$

Density = $D = M/V$



Mass of dust = $m = fM$

Area of dust = $A = 4\pi R^2$

Surface density = $S = m/A$

Fourth gather any information that might be useful:

Know:

$D = 2.0 \text{ g/cm}^3$

$f = 0.05$

$R = 6380 \text{ km}$

$S = 0.020 \text{ g/cm}^2$

Unknown:

M

r

V

m

A

Fifth, specify the quantity that you wish to calculate:

Target quantity is r, the radius of the asteroid

Sixth, outline the process by which you can calculate the quantity you want.

Identify an equation that has your target quantity. Identify the other unknowns in that equation. Choose one of those unknowns and choose an equation that you have not yet chosen that has that unknown. Continue this process until you have a different equation matched to each unknown.

	unknowns
Find r	r
$V = (4/3)\pi r^3$	V
Find V	
$D = M/V$	M
Find M	
$m = fM$	m
Find m	
$S = m/A$	A
Find A	
$A = 4\pi R^2$	

Seventh, follow your process by working from the last step to the first step which is your target. Do not put in numbers until you are sure your calculation works:

Solve for r by going from the last equation to the first

$$\begin{aligned}
 A &= 4\pi R^2 \\
 S &= m/(4\pi R^2) \\
 m &= S \ 4\pi R^2 \\
 (S \ 4\pi R^2) &= f \ M \\
 (S \ 4\pi R^2)/f &= M \\
 D &= (S \ 4\pi R^2/f)/V \\
 V &= (S \ 4\pi R^2)/(f \ D) \\
 (S \ 4\pi R^2)/(f \ D) &= (4/3)\pi r^3 \\
 r^3 &= 3 \ S \ R^2/(f \ D)
 \end{aligned}$$

Eighth, check that the units of your answer are correct to determine if you have made an algebra mistake. If you have, go back and correct it:

$$\text{Check units: } [m^3] = [g/m^2] [m^2]/[g/m^3] = [m^3]$$

The units of both side of the equation have units of mass so the units check.

Ninth, put in numbers and calculate a numerical answer:

$$r = \sqrt[3]{\frac{3SR^2}{fD}}$$

$$r = \sqrt[3]{\frac{3\left(0.02\frac{\text{g}}{\text{cm}^2}\right)(6380\text{km})^2}{(0.05)\left(2.0\frac{\text{g}}{\text{cm}^3}\right)}}$$

$$r = \sqrt[3]{\frac{2.4 \times 10^6 \left(\frac{\text{g}}{\text{cm}^2}\right)(\text{km})^2}{\left(0.1\frac{\text{g}}{\text{cm}^3}\right)}}$$

$$r = \sqrt[3]{2.4 \times 10^7 (\text{cm})(\text{km})^2}$$

$$r = \sqrt[3]{2.4 \times 10^7 (10^{-5}\text{km})(\text{km})^2}$$

$$r = \sqrt[3]{2.4 \times 10^2 (\text{km})^3}$$

$$r = 6.2 \text{ km} \approx 4 \text{ miles}$$

Tenth, check that the units of your answer are correct to determine if you have made a numerical mistake. If you have, go back and correct it:

The units of the radius is a distance so the units are correct.

Eleventh, check that you have actually answered the question and that your answer makes sense. Make sure it fits in with something you know. If not, go back, find

Evaluation:

The question was answered. The radius is a measure of the size of an object. (Note that you also could have equally well have answered the question by calculating the asteroid's diameter, volume, or total mass.)

An asteroid is much smaller than the size of the Earth (8000 miles). Which the answer is. Also, I know that the diameter of a medium size asteroid is about 10 miles so a 4 mile diameter asteroid is on the small size but not unreasonable.